

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

With rise in temperature, the Young's modulus of elasticity

- (1) changes erratically
- (2) decreases
- (3) increases
- (4) remains unchanged

Q2 - 2024 (01 Feb Shift 2)

One end of a metal wire is fixed to a ceiling and a load of 2 kg hangs from the other end. A similar wire is attached to the bottom of the load and another load of 1 kg hangs from this lower wire. Then the ratio of longitudinal strain of upper wire to that of the lower wire will be _____

[Area of cross section of wire = 0.005 cm^2 , $Y = 2 \times 10^{11} \text{ Nm}^{-2}$ and $g = 10 \text{ ms}^{-2}$]

Q3 - 2024 (27 Jan Shift 1)

If average depth of an ocean is 4000 m and the bulk modulus of water is $2 \times 10^9 \text{ Nm}^{-2}$, then fractional compression $\frac{\Delta V}{V}$ of water at the bottom of ocean is $\alpha \times 10^{-2}$. The value of α is _____ (Given, $g = 10 \text{ ms}^{-2}$, $\rho = 1000 \text{ kg m}^{-3}$)

Q4 - 2024 (27 Jan Shift 2)

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : The property of body, by virtue of which it tends to regain its original shape when the external force is removed, is Elasticity.

Reason (R) : The restoring force depends upon the bonded inter atomic and inter molecular force of solid.

In the light of the above statements, choose the correct answer from the options given below :

- (1) (A) is false but (R) is true
- (2) (A) is true but (R) is false
- (3) Both (A) and (R) are true and (R) is the correct explanation (A)

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(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

Q5 - 2024 (29 Jan Shift 2)

A wire of length L and radius r is clamped at one end. If its other end is pulled by a force F , its length

increases by l . If the radius of the wire and the applied force both are reduced to half of their original values keeping original length constant, the increase in length will become.

(1) 3 times

(2) $3/2$ times

(3) 4 times

(4) 2 times

Q6 - 2024 (29 Jan Shift 2)

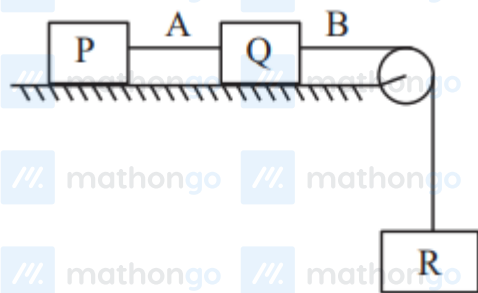
Two metallic wires P and Q have same volume and are made up of same material. If their area of cross sections are in the ratio 4 : 1 and force F_1 is applied to P, an extension of Δl is produced. The force which is required to produce same extension in Q is F_2 .

The value of $\frac{F_1}{F_2}$ is _____

Q7 - 2024 (30 Jan Shift 1)

Each of three blocks P, Q and R shown in figure has a mass of 3 kg. Each of the wire A and B has cross-sectional area 0.005 cm^2 and Young's modulus $2 \times 10^{11} \text{ N m}^{-2}$. Neglecting friction, the longitudinal strain on

wire B is $\times 10^{-4}$. (Take $g = 10 \text{ m/s}^2$)



Q8 - 2024 (31 Jan Shift 1)

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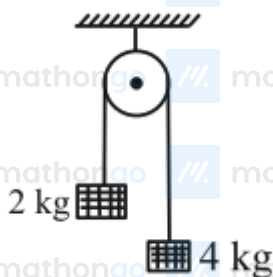
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The depth below the surface of sea to which a rubber ball be taken so as to decrease its volume by 0.02% is _____ m.

(Take density of sea water = 10^3 kg m^{-3} , Bulk modulus of rubber = $9 \times 10^8 \text{ N m}^{-2}$, and $g = 10 \text{ m s}^{-2}$)

Q9 - 2024 (31 Jan Shift 2)

Two blocks of mass 2 kg and 4 kg are connected by a metal wire going over a smooth pulley as shown in figure. The radius of wire is $4.0 \times 10^{-5} \text{ m}$ and Young's modulus of the metal is $2.0 \times 10^{11} \text{ N/m}^2$. The longitudinal strain developed in the wire is $\frac{1}{\alpha\pi}$. The value of α is _____. [Use $g = 10 \text{ m/s}^2$]



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Q1 (2) mathongo /// math Q**2** (3) /// math **Q3** (2) math mc **Q4** (3) /// math

Q5 (4) math /// math **Q6** (16) /// math **Q7** (2) math mc **Q8** (18) /// math

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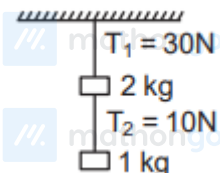
Solutions

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Q1

Conceptual questions

Q2



$$\Delta L = \frac{FL}{AY}$$

$$\frac{\Delta L}{L} = \frac{F}{AY}$$

$$\frac{\frac{\Delta L_1}{L_1}}{\frac{\Delta L_2}{L_2}} = \frac{F_1}{F_2} = \frac{30}{10} = 3$$

Q3

$$B = -\frac{\Delta P}{\left(\frac{\Delta V}{V}\right)}$$

$$-\left(\frac{\Delta V}{V}\right) = \frac{\rho gh}{B} = \frac{1000 \times 10 \times 4000}{2 \times 10^9}$$

$$= 2 \times 10^{-2} \text{ [-ve sign represent compression]}$$

Q4

Theory

Q5

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Solutions

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$$Y = \frac{\text{stress}}{\text{strain}}$$

$$Y = \frac{\frac{F}{\pi r^2}}{\frac{\Delta \ell}{\ell}}$$

$$F = Y \pi r^2 \times \frac{\Delta \ell}{\ell} \dots (i)$$

$$Y = \frac{\frac{F r^2}{2}}{\frac{\Delta \ell}{\ell}}$$

$$F = Y \frac{\Delta \ell}{\ell} \times 2 \times \frac{\pi r^2}{4}$$

$$\text{From (i)}$$

$$Y \pi r^2 \frac{\ell}{L} = Y \frac{\Delta \ell}{L} \frac{\pi r^2}{2}$$

$$\Delta \ell = 2\ell$$

$$\Delta \ell = 2\ell$$

Q6

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{\Delta \ell / \ell} = \frac{F \ell}{A \Delta \ell}$$

$$\Delta \ell = \frac{F \ell}{A Y}$$

$$V = A \ell \Rightarrow \ell = \frac{V}{A}$$

$$\Delta \ell = \frac{F V}{A^2 Y}$$

Y & V is same for both the wires

$$\Delta \ell \propto \frac{F}{A^2}$$

$$\frac{\Delta \ell_1}{\Delta \ell_2} = \frac{F_1}{A_1^2} \times \frac{A_2^2}{F_2}$$

$$\Delta \ell_1 = \Delta \ell_2$$

$$F_1 A_2^2 = F_2 A_1^2$$

$$\frac{F_1}{F_2} = \frac{A_1^2}{A_2^2} = \left(\frac{4}{1}\right)^2 = 16$$

$$\frac{F_1}{F_2} = \frac{A_1^2}{A_2^2} = \left(\frac{4}{1}\right)^2 = 16$$

Q7

$$\Delta \ell = \frac{F \ell}{A Y}$$

$$\Delta \ell = \frac{F \ell}{A Y}$$

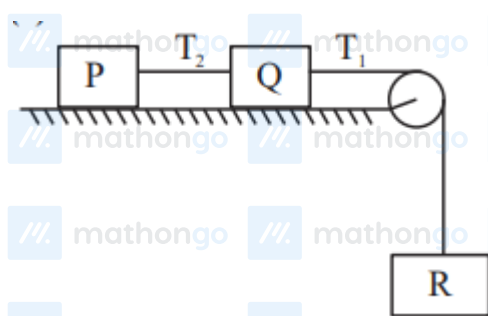
$$\Delta \ell = \frac{F \ell}{A Y}$$

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Solutions

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$$a = \frac{10}{3} \text{ m/s}^2$$

$$30 - T_1 = 3 \times a$$

$$T_1 = 20 \text{ N}$$

$$\text{strain} = \frac{\text{stress}}{Y}$$

$$= 2 \times 10^{-4}$$

Q8

$$\beta = \frac{-\Delta P}{\frac{\Delta V}{V}}$$

$$\Delta P = -\beta \frac{\Delta V}{V}$$

$$\rho g h = -\beta \frac{\Delta V}{V}$$

$$10^3 \times 10 \times h = -9 \times 10^8 \times \left(-\frac{0.02}{100} \right)$$

$$\Rightarrow h = 18 \text{ m}$$

Q9

$$T = \left(\frac{2 m_1 m_2}{m_1 + m_2} \right) g = \frac{80}{3} \text{ N}$$

$$A = \pi r^2 = 16\pi \times 10^{-10} \text{ m}^2$$

$$\text{Strain} = \frac{\Delta \ell}{\ell} = \frac{F}{AY} = \frac{T}{AY}$$

$$= \frac{80/3}{16\pi \times 10^{-10} \times 2 \times 10^{11}} = \frac{1}{12\pi}$$

$$\alpha = 12$$

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